

HT PBPK Prediction Framework documentation

ESQlabs

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1 Technical requirements

- Windows operating system (Linux not tested)
- R4.3x
- git (for installing packages) - <https://git-scm.com/download/win>
- R packages:
 - {esqlabsR} - <https://github.com/esqLABS/esqlabsR>

- {uuid} - CRAN
- {pracma} - CRAN
- {readr} - CRAN

2 Installation

Install the latest version of the `{esqlabsR}` package and the dependencies by calling the following code:

```
install.packages("pak")
pak::pak("esqlabs/esqlabsR@*release")
```

Install the remaining package from CRAN by calling the following code:

```
install.packages(c(
  "uuid", "pracma",
  "readr"
))
```

3 Workflow

The high-throughput (HT) physiologically-based (PB) pharmacokinetics (PB) modeling framework is distributed as an R-Studio project with the required folder structure. For the correct execution of the implemented code, the working directory must be the `Code` subfolder.

The framework requires the in vitro/in silico compound properties and the in vivo PK data to be present in an Excel file. The name and the location of the file is defined in the `ProjectConfiguration.xlsx` located in the root project folder.

3.1 Current framework limitations

- Only simulations of a small molecule structure are supported
- Only simulations of the rat species are supported
- Only single dose intravenous administration protocols are implemented

3.2 Running simulations

- 1) Initialization of the framework is done by executing the script `HTFramework.R`. It must be called once in an R session before running the predictions. Within the script file, the scenarios (representing the different calculation methods) can be specified. Each study will be simulated using every specified scenario. The script calls the function `initHTFramework()`, which loads the simulations from the `'pkml'` files and returns the initialized scenarios.

Simulation outputs can be defined in the `Scenarios.xlsx` file (see workflow description <https://esqlabs.github.io/esqlabsR/articles/esqlabsR-design-scenarios.html>).

- 2) After the framework has been initialized, the function `createStudies()` is called to create a **Study** object for every study defined in the input data file. The function returns a list of **Study** objects. These objects can be viewed and modified.
- 3) Calling the function `runPredictions()` with the list of **Study** objects and the list of scenarios will run the simulations. The function will create and process a simulation for every study and every scenario.

4 Implementation details

4.1 Generic PBPK model structure

4.1.1 Individual

A generic PBPK model is implemented in PK-Sim version 11.3. The project includes the species human, rat, dog, mouse, minipig, and monkey. This version of the HT-PBPK framework considers only the rat species. The simulation time will be set to the maximal measured (reported in the column **TIME** of the **EXP_PK** sheet) plus 10%. The maximal measured value in the provided exemplary data set is 24 hours.

4.1.2 Compound

A generic compound named **Compound** is used for simulations. The distinction of small vs. large molecule will be defined in the R code by setting the value of the parameter `Compound|Is small molecule`. In the first version of the framework, only small molecules are simulated.

The lipophilicity, the fraction unbound, of the compound will be set in the R script from the PhysChem data provided. As plasma binding partner, Albumin is assumed by default.

For solubility, one value at a given pH (by default 7) will be provided. Solubility table (multiple solubility values at different pH) is not supported in this framework.

Intestinal permeability will be calculated by PK-Sim unless a specific value is provided.

Clearance will be implemented as two processes - glomerular filtration in the kidney and the liver plasma clearance. For the liver clearance process, **In vitro liver microsomes** - **t1/2** will be used. For the glomerular filtration, a filtration fraction of 1 and 0 will be assumed for small and large molecules, respectively.

No proteins - transporter or enzymes - are considered.

4.1.3 Administration protocols

Intravenous bolus, oral solution, and subcutaneous (administered into skin interstitial) will be implemented. For oral administration, solution will be assumed as the only available formulation.

No food administration or other events will be implemented.

i Note

Current implementation supports SD IV administration protocols only.

4.1.4 Calculation methods

For each combination of the partition coefficients (PC) and cellular permeabilities (CP) calculation methods implemented in PK-Sim, a separate simulation will be created. Additionally, separate simulations for the small molecules (SM) and large molecules (LM) model structures will be created. The combinations of PC/CP calculation methods are (in parenthesis - abbreviation used in output file names):

- PK-Sim Standard / PK-Sim Standard (PKSim_PKSim)
- PK-Sim Standard / Charge dependent Schmitt (PKSim_Schmitt)
- Rodgers and Rowland / PK-Sim Standard (RR_PKSim)
- Rodgers and Rowland / Charge dependent Schmitt (RR_Schmitt)
- Schmitt / PK-Sim Standard (Schmitt_PKSim)
- Schmitt / Charge dependent Schmitt (Schmitt_Schmitt)
- Poulin and Theil / PK-Sim Standard (PT_PKSim)
- Poulin and Theil / Charge dependent Schmitt (PT_Schmitt)
- Berezhkovskiy / PK-Sim Standard (Berezh_PKSim)
- Berezhkovskiy / Charge dependent Schmitt (Berezh_Schmitt)

Note: CP calculation method “Charge dependent Schmitt normalized to PK-Sim” is not supported as it is not properly implemented in the current versions (11.3 and 12.0) of PK-Sim.

4.2 Input data structure

The input data (i.e., compound properties and PK data) are provided in an Excel file with the sheets `Compound_Properties`, `ADME_Properties`, and `EXP_PK`.

Each compound is defined by a unique identifier (ID) in the column `COMPOUND_ID`.

4.2.1 Compound_Properties

Physico-chemical (PhysChem) properties of the compound are provided in the sheet `Compound_Properties`. Available properties are defined in the column `PREDICTION_ENDPOINT`. The following properties are available:

```
unique(dataCompoundProperties$PREDICTION_ENDPOINT)
```

[1] "Acidic_pKa"	"Basic_pKa"	"Caco2-TC7_Permeability"
[4] "F"	"MW"	"Solubility"
[7] "logD_Lipophilicity"		

The data will be mapped to model parameters as follows:

- MW: Compound|Molecular weight
- F, Br, Cl, I: Number of halogens (Br, Cl, F, I). 0 if not available.
- logD_Lipophilicity: Compound|Lipophilicity
- Acidic_PkA, Basic_PkA: pKa values of the molecule. It is assumed that up to three entries exist per compound. If more than three entries exist, only the first three will be applied.
- Caco2-TC7_Permeability: Compound|Specific intestinal permeability (transcellular). If not available, the value is calculated by PK-Sim based on the molecular weight and the lipophilicity of the compound.
- Solubility: Compound|Solubility at reference pH

It is assumed that all values are provided in units (defined in the column `RESULT_UNIT`) that are compatible with PK-Sim. Following units are available:

```
unique(dataCompoundProperties$RESULT_UNIT)
```

```
[1] "no unit" "cm/s"      "g/mol"      "μmol/l"
```

4.2.2 ADME_Properties

Species-specific compound properties are provided in the tab `ADME_Properties`. Only entries with `SPECIES == Rat` will be used by the framework.

The following properties are available:

```
unique(dataADMEProperties$PREDICTION_ENDPOINT)
```

```
[1] "Microsomal_Half_Life_Full"  "Microsomal_Half_Life_Single"
[3] "Plasma_Protein_Binding"
```

The data will be mapped to model parameters as follows:

- Microsomal_Half_Life_Full: Compound-Total Hepatic Clearance-In vitro microsomes Rat|t_{1/2} (microsomal assay)
- Microsomal_Half_Life_Single: Only used if Microsomal_Half_Life_Full is not available.
- Plasma_Protein_Binding: Fraction bound to plasma proteins (Albumin assumed).

It is assumed that plasma protein binding is provided for each compound.

4.2.3 Study protocols

Study protocols are provided within the `EXP_PK` sheet. A separate simulation will be executed for each study defined in the `STUDY` column. Only studies for compounds that are defined in the sheets `Compound_Properties` and `ADME_Properties` and with `SPECIES == Rat` will be simulated.

Administration protocols will be defined in the following columns:

- `TREATMENT_ROUTE` with options:
 - IV bolus
 - PO
 - SC
- `TREATMENT_REGIMEN` with options:
 - SD (single administration only, given at time = 0)
 - QD (once daily, given every 24h)
 - BID (twice daily, given every 8h)
 - QW (once weekly, given every 7 days)
 - Q2W (once every two weeks, given every 14 days)
 - Q4W (once every four weeks, given every 28 days)
- `TREATMENT_REPS` - numeric, total number of repetitions

i Note

Currently not implemented.

For each unique study ID, one simulation will be performed, i.e., no individual simulations will be executed in case results for multiple individuals for the same study are available. The first entry in columns `TREATMENTDESC` and `DOSE` will be used for treatment parametrization.

4.3 In vitro data pre-processing

- Solubility is converted from molar concentration to mass concentration using the molecular weight of the compound.
- Unit values `no unit` will be converted to “.”.
- All values are converted to [default OSPS units](#)
- Remove compounds that have no plasma protein binding information
- PK-Sim models require plasma protein binding to be under 100%. For all compounds with reported plasma protein binding of $\geq 100\%$, replace the value by 99.9%. Current values are:

```
summary(dataADMEProperties$RESULT_NUMERIC[dataADMEProperties$PREDICTION_ENDPOINT == "Plasma_Prote
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
40.82	41.66	42.50	42.50	43.34	44.18

4.4 In vivo data pre-processing

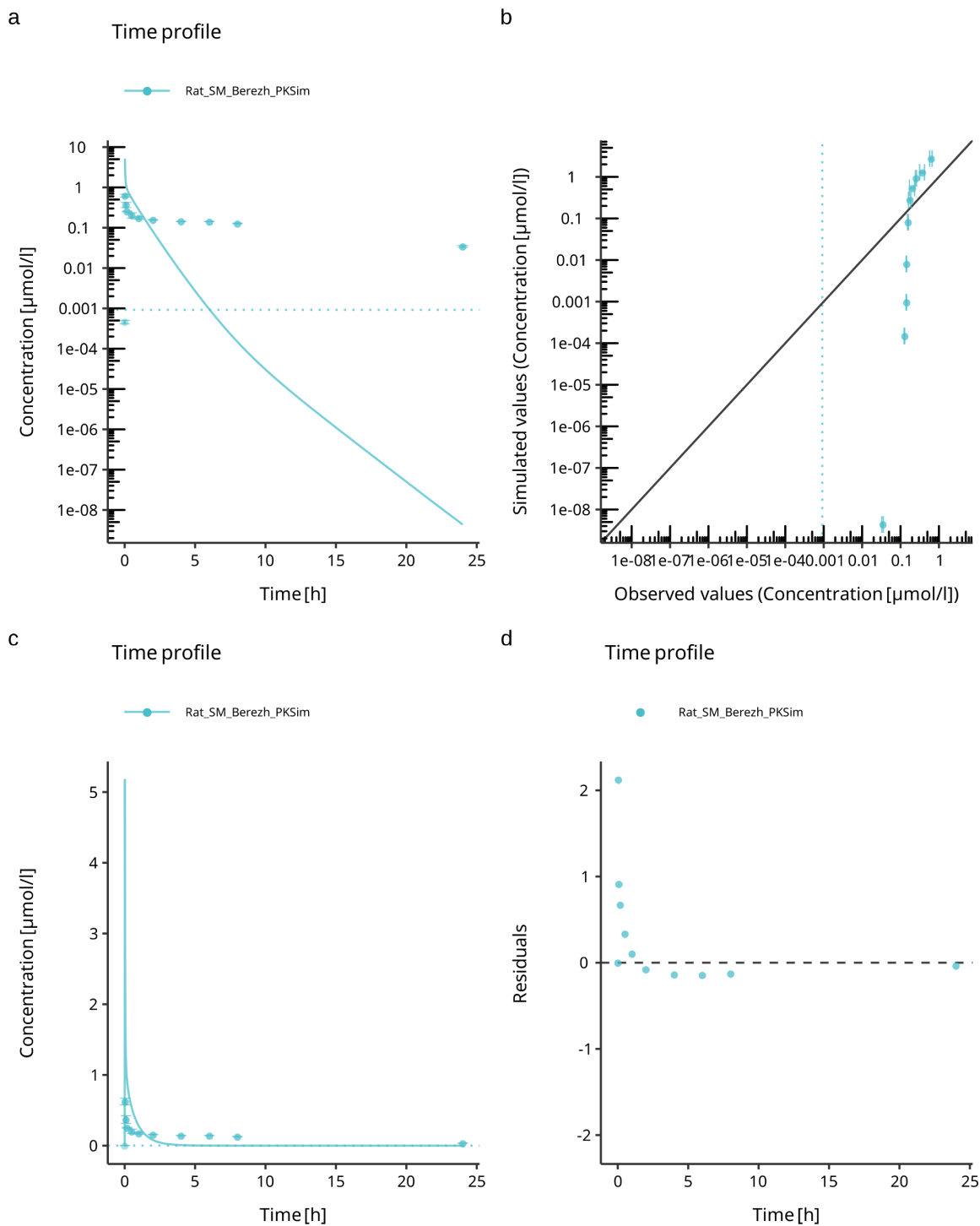
Duplicated time entries of the observed data are removed.

5 Framework outputs

Each time the function `runPredictions()` is called, the framework generates the following outputs:

- A folder with the name `<timestamp>` located in the subfolder `SimulationResults` of the output folder defined in the `ProjectConfiguration.xlsx` (by default the `Results` folder of the project).
- Within the `<timestamp>` folder, subfolder structure `<CompoundID>/<StudyID>` is created.
- For each simulated study:
 - For each `Scenario` (defined by the variable `scenarioNames` in the script `HTFramework.R`), which represents a combination of PC and CP calculation methods:
 - Simulated results as a `scv` format file, the file name of the form `<CompoundID>_<StudyID>_<Species>_SM_<` where `SM` stands for “Small molecule” (`LM` for “Large molecule”), `<PC>` and `<CP>` are the partition coefficient and cellular permeability calculation methods, respectively.
 - If `saveSimulation == TRUE`, the parametrized simulation as `.pkml` file.
 - PK parameters as calculated by the standard function `calculatePKAnalyses()` implemented in the `{ospsuite}` package, stored as `csv` with the file name `<CompoundID>_<StudyID>_<Species>_SM_<PC`
 - If `plotFigures == TRUE`, a multi-panel plot of the type:

CMP_ID_1_STUDY_ID_4_Rat_SM_Berezh_PKSim



- A **csv** file with the combined metrics calculated for the simulation results of all studies and scenarios. The file is named **Prediction_metrics.csv**, containing the following columns:

- **obsFirstTime**: Time of the first observed data point
- **obsFirstConc**: Concentration of the first observed data point
- **simFirstConc**: Concentration of the first simulated data point
- **obsLastTime**: Time of the last observed data point
- **obsLastConc**: Concentration of the last observed data point
- **simLastConc**: Concentration of the last simulated data point
- **foldC_first**: Fold difference between the first simulated and observed data points, calculated as $\frac{simFirstConc}{obsFirstConc}$
- **foldC_last**: Fold difference between the last simulated and observed data points, calculated as $\frac{simLastConc}{obsLastConc}$
- **obsCmax**: Maximum observed concentration
- **simCmax**: Maximum simulated concentration
- **cMaxFold**: Fold difference between the maximum simulated and observed concentrations, calculated as $\frac{simCmax}{obsCmax}$
- **log residuals**: $\ln(simulated) - \ln(observed)$
- **lin residuals**: $simulated - observed$
- **observed_AUC**: Calculated for time points present in the observed data using function `trapz()` from package `{pracma}`.
- **simulated_AUC**: Calculated for time points present in the simulated data using function `trapz()` from package `{pracma}`.
 - * **AUCfold**: Fold difference between the simulated and observed AUCs, calculated as $\frac{simulated_AUC}{observed_AUC}$
- **fold difference**: $\frac{max(observed, simulated)}{min(observed, simulated)}$ for each observed time point
- **GMFE**: calculated as $e^{mean(abs(log_residuals))}$
- **RMSE**: calculated as $\sqrt{mean(lin_residuals^2)}$
- **RMedianSE**: calculated as $\sqrt{median(log_residuals^2)}$
- **share_x_fold**: % of points within x fold. Calculated for 1.5, 2, 3, 5, 10 folds.